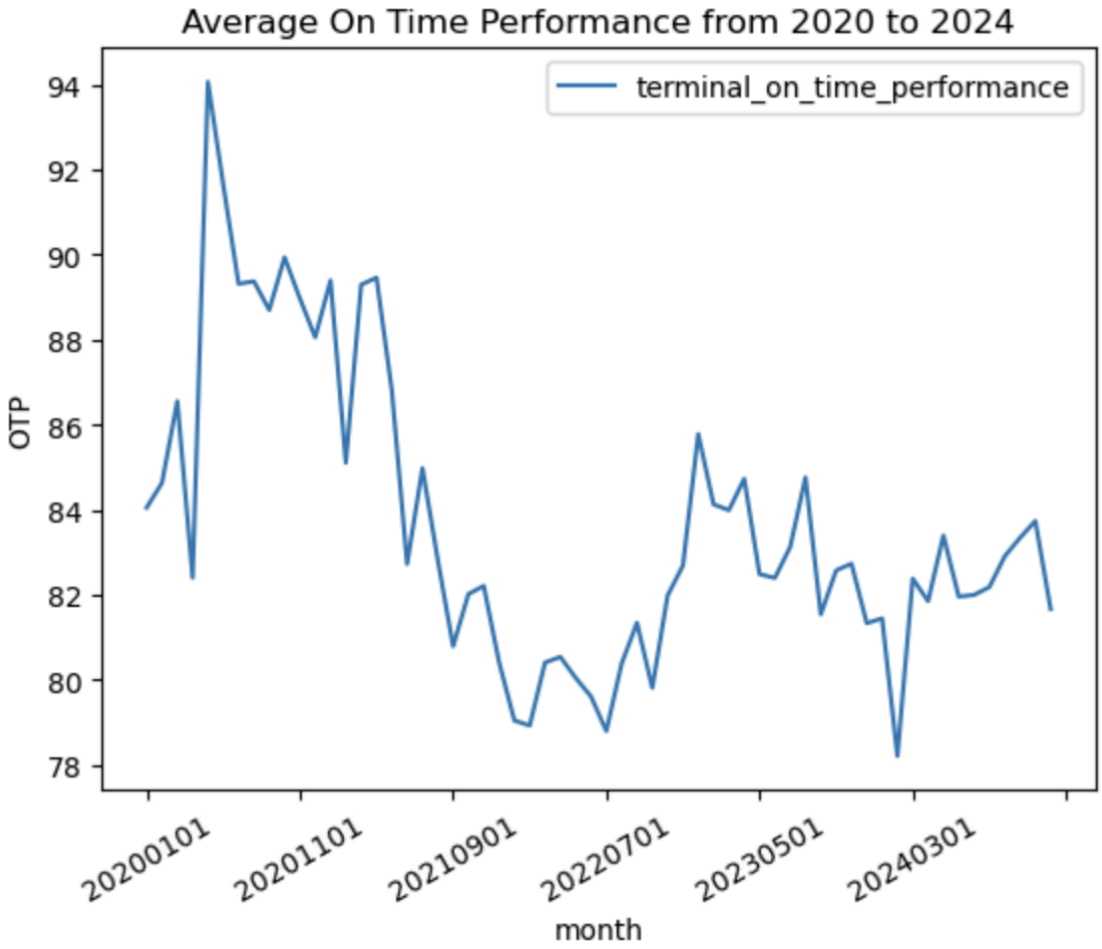
How does Subway On Time Performance Vary by Line and Day Type from 2020 to 2024?



Introduction

The NYC subway is one of the most extensive systems in the world. There are 25 lines which are split up by division (A for numbered lines and B for lettered), which avenues they run on, and if they run express or local. Subway performance is measured by OTP (On Time Performance). According to the MTA website, OTP is the percentage of lines that arrive at their final stop within 5 minutes and 59 seconds of the scheduled time. The OTP is important because if subways aren't running efficiently or on schedule, it can cause people to be late for jobs, classes, or events. It can also be bad for people who take elevated lines if someone has to wait a long time in severe heat or cold. This project will show how well different lines perform.

Data Methods

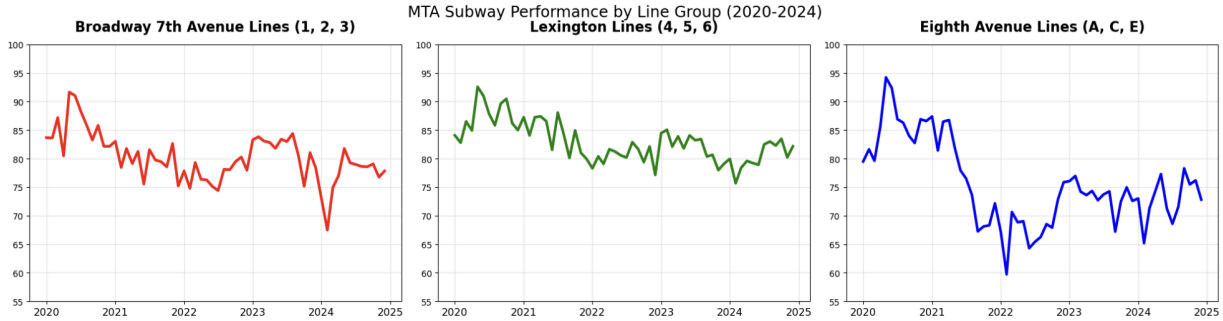
- Downloaded On Time Performance CSV from MTA Open Data.
- The data file had each month from 2020 to 2024, if it was weekend or weekday, which line it was, the scheduled number of trips, the actual number of arrivals, as well as the OTP.
- Cleaned up data to get rid of extra symbols (-, %, ,).
- Uploaded cleaned file to Jupyter Lab.
- Used pandas to separate data into categories (express, local, shuttle for each day type)
- Created 5 tiers to analyze performance (below 75 percent, between 75 and 80, between 80 and 85, between 85 and 90, as well as above 90).
- Printed the top and bottom 5 from each division and tier.
- Created Line and bar charts with colors that represent the subway line colors.

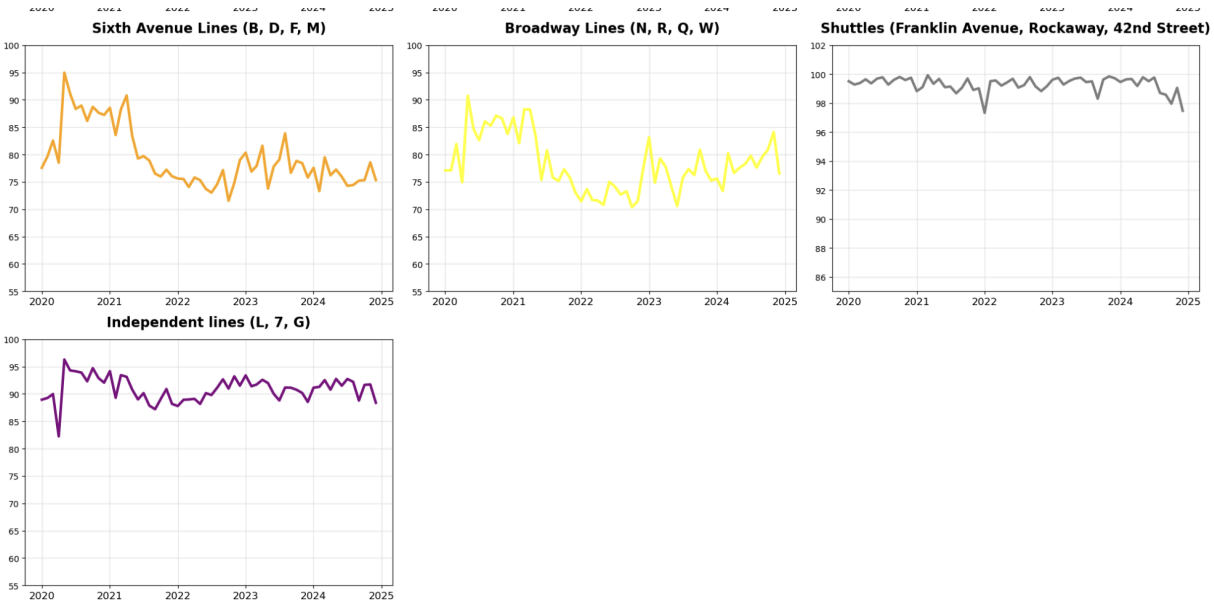
Code Link

https://github.com/joshdunford03/Subway-OTP-Performance-2020-to-2024/blob/main/transitdata.ipynb

Figure 1

Average OTP for Each Line Group





<u>Line Guide</u>

A Division - Numbered Lines
B Division - Lettered Lines

<u>Shuttles</u>
Manhattan Shuttle - Grand Central to Times Square
Queens Shuttle - Rockaway Park 116th Street to Broad Channel
Brooklyn Shuttle - Franklin Avenue to Prospect Park

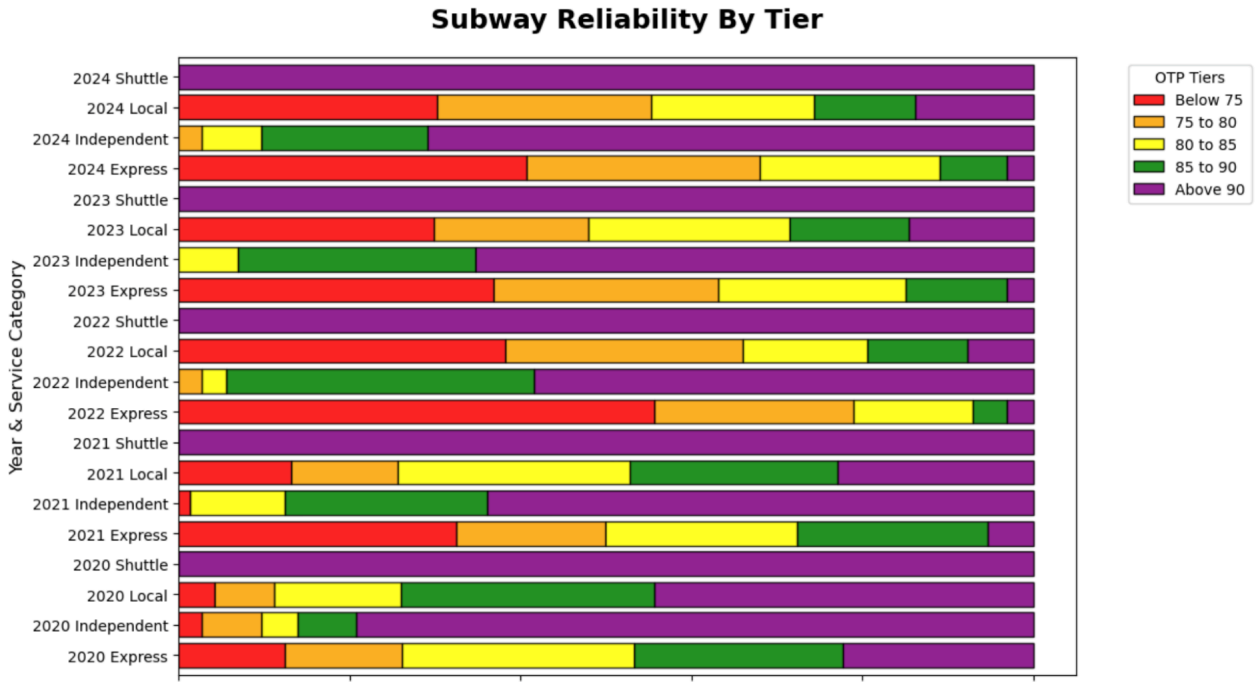
<u>Independent Lines</u> (don't share tracks with any other trains)
7 (local) - Flushing Main Street to 34th Street Hudson Yards
G (local) - Court Square to Church Avenue
L (local) - L - 8th Avenue to Rockaway Parkway

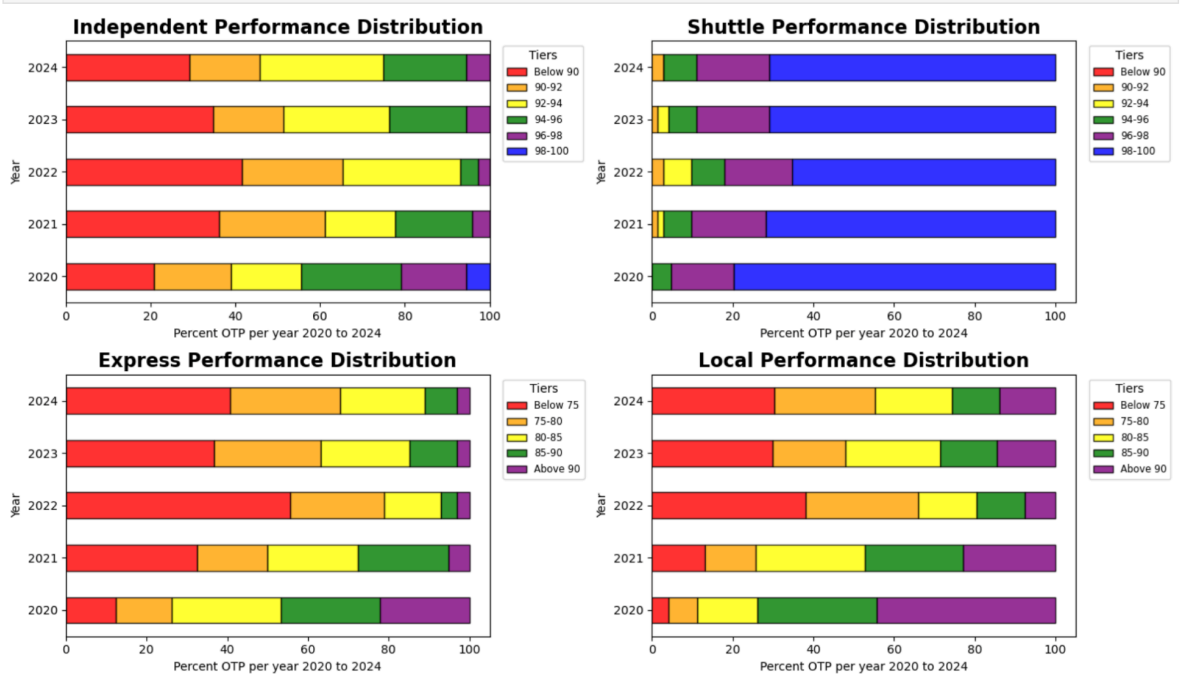
<u>Local</u>
1 - Van Cortlandt Park to South Ferry
6 - Pelham Bay Park to Brooklyn Bridge City Hall
C - 168th Street to Euclid Avenue
R - Forest Hills 71st Avenue to Bay Ridge 95th Street
W -Astoria Ditmars to Whitehall Street - South Ferry
F - Jamaica 179th Street to Coney Island)
M - Forest Hills 71st Avenue to Middle Village-Metropolitan Avenue
J - Broad Street to Jamaica Center Parsons/Archer
Z - Jamaica Center-Parsons/Archer to Broad Street

<u>Express</u>

2 - Wakefield 241st Street to Flatbush Avenue/Brooklyn College
3 - Harlem 148th Street to New Lots Avenue
4 - Woodlawn to Crown Heights Utica Avenue
5 - Eastchester Dyre Avenue to Flatbush Avenue Brooklyn College
A - Inwood 207th Street to Rockaway Park
B - Bedford Park Boulevard to Brighton Beach
D - Norwood 205th Street to Coney Island
E - Jamaica Center-Parsons Archer to World Trade Center
N - Astoria Ditmars to Coney Island
Q - E 96th Street to Coney Island

Figure 2 and 3


Subway Reliability By Tier
Year & Service Category
2024 Shuttle
2024 Local
2024 Independent
2024 Express
2023 Shuttle
2023 Local
2023 Independent
2023 Express
2022 Shuttle
2022 Local
2022 Independent
2022 Express
2021 Shuttle
2021 Local
2021 Independent
2021 Express
2020 Shuttle
2020 Local
2020 Independent
2020 Express
OTP Tiers
Below 75
75 to 80
80 to 85
85 to 90
Above 90


Independent Performance Distribution
Shuttle Performance Distribution
Express Performance Distribution
Local Performance Distribution
Year
2024
2023
2022
2021
2020
Percent OTP per year 2020 to 2024
Tiers
Below 90
90-92
92-94
94-96
96-98
98-100
Below 75
75-80
80-85
85-90
Above 90

Top and Bottom 5 Routes in Each Division, Day Type, and Tier

Highest performing Express (Above 90)

Weekday A Division

Month	Line	OTP
May 2020	4	94.32
May 2020	3	93.67
June 2020	3	93.47
April 2020	4	92.70
May 2020	2	92.55

Weekday B Division

Month	Line	OTP
May 2020	E	96.88
May 2020	Q	95.98
June 2020	E	94.01
May 2020	D	93.96
May 2020	A	92.99

Weekend A Division

Month	Line	OTP
October 2020	5	99.01
September 2022	5	96.78
June 2020	5	96.46
March 2023	5	95.22
May 2020	5	94.84

Weekend B Division

Month	Line	OTP
March 2021	E	98.15
May 2020	E	96.44
January 2021	E	95.75
August 2020	D	95.73
June 2020	A	94.75

<u>Lowest performing express</u>

Weekday A Division

Month	Line	OTP
January 2024	2	69.90
December 2023	2	69.60
October 2023	2	68.56
September 2022	2	68.09
December 2024	2	69.07

Weekday B Division

Month	Line	OTP
November 2022	B	59.87
December 2024	B	59.76
August 2024	B	54.17
September 2024	B	52.85
October 2022	B	51.21

Weekend A Division

Month	Line	OTP

April 2021	2	54.76
May 2022	2	53.45
April 2020	2	51.62
June 2021	2	49.29
February 2024	2	31.01

Weekend B Division

Month	Line	OTP
June 2022	A	54.77
July 2024	A	54.19
August 2022	D	53.18
April 2020	N	51.34
February 2022	A	50.64

<u>Highest performing local</u>

Weekday A Division

Month	Line	OTP
May 2020	7	98.63
May 2020	1	97.21
October 2020	7	96.90
June 2020	1	96.62
September 2020	7	96.05

Weekday B Division

Month	Line	OTP
May 2020	L	98.85
May 2020	M	97.84

June 2020	L	97.10
May 2020	G	96.33
April 2021	L	96.09

Weekend A Division

Month	Line	OTP
August 2020	7	97.57
January 2021	7	97.44
July 2020	1	96.79
July 2020	7	96.76
March 2024	1	96.75

Weekend B Division

Month	Line	OTP
March 2020	L	99.40
October 2020	M	99.18
November 2020	M	99.02
September 2020	M	99.02
January 2021	M	99.00

<u>Lowest performing local</u>

Weekday A Division

Month	Line	OTP
January 2022	6	73.97
December 2023	6	72.89
January 2024	1	72.73

November 2023	6	72.48
March 2024	6	67.83

Weekday B Division

Month	Line	OTP
December 2024	F	63.95
September 2021	C	63.43
September 2023	F	61.42
January 2024	F	60.16
February 2024	F	54.84

Weekend A Division

Month	Line	OTP
June 2024	6	72.79
May 2022	6	71.40
April 2020	6	71.28
April 2020	7	64.50
December 2022	6	63.85

Weekend B Division

Month	Line	OTP
May 2022	R	51.27
June 2022	C	50.54
February 2024	C	49.00
July 2024	C	47.09
February 2022	C	41.03

<u>Key Findings</u>

Top performing lines all 4 years

- The 7, L, and G had the highest OTP for each year but 7 had low OTP in April 2020
- Shuttles also had very high OTP levels each year.
- The 7, L, and G performed well because of modern signalling.
- These lines as well as the shuttles use their own tracks and so their performances were higher than trains that share track lines.

Low OTPs

- The lowest OTP was the 2 train in February 2024 (31 percent). This is one of the longest routes in the city, going from Wakefield 241st in the Bronx to Flatbush Avenue.
- Longer routes like the 2 and A could have lower OTPs because of how many stops they have
- Some lines had OTPs in the 40 - 50 percent range. These are way below 75 percent and the low OTP could have been caused by track sharing, long dwell times at specific stations, or signal errors.
- Lines that share tracks can be delayed if one train is malfunctioning (kind of like a chain effect that impacts the other lines).
- Local lines had higher OTP levels than express some months despite express lines making fewer stops.

Better Performance in 2020

- All lines performed better during the Covid pandemic. Low ridership reduced the time at each station since if nobody was on the tracks, the train could leave faster.
- The ridership levels decreased significantly during the pandemic which negatively impacted fare revenue but gave the MTA more opportunities for track work.

<u>Next Steps</u>

- See how data has changed since the end of 2024.
- Connect to how summer and winter OTPs can be dangerous if stations don't have air conditioning during summer months or heat in winter.
- See how each station on the branches were affected (the 2 train to Brooklyn College having 31 percent may have negatively impacted students trying to get to class).
- Perform the same analysis on bus routes in different boroughs since they are planned differently due to street design and terrain changing by borough.
- Compare slow bus routes in outer boroughs and Manhattan to high performing routes like the M15 SBS to see how dedicated bus lanes, stop spacing, and all door boarding leads to better performance.

<u>Conclusion</u>

Subway OTP varies a lot by the type of train, the way the tracks are planned, and if they use new signalling technology. The highest performing lines were the 7, L, and G which use new signalling and dedicated tracks. The MTA is currently working on improving signals across all lines which will increase the OTP levels by a lot. Despite the OTP issues and some lines performing slower, the subway still moves millions of people throughout the city each day and is one of the most integral parts of the city.

Data Source for Analysis

MTA Subway Terminal On-Time Performance 2020 - 2024
https://data.ny.gov/Transportation/MTA-Subway-Terminal-On-Time-Performance-2020-2024/vtvh-gimj/about_data